Improved NP-Hardness of Approximation for Max-3-Cut and for Max-k-Cut with Small k

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Abstract

We prove that Max-3-Cut is NP-hard to approximate within a factor of $27/29 + \varepsilon \approx 0.931$. This improves the factor $16/17 \approx 0.941$ of Austrin, O'Donnell, Tan and Wright, the best NP-hardness result for Max-3-Cut since 2012. (Their result, unlike ours, also holds for 3-colourable graphs.) We also improve the NP-hardness of Max-k-Cut for every k from 4 to 20; for instance, we obtain $45/47 \approx 0.957$ for Max-4-Cut and $33/34 \approx 0.971$ for Max-5-Cut, where the previous factors were 0.987 and 120/121.

Our reductions are gadget reductions from Håstad's linear equations over $\mathbb{Z}_3$, with gadgets that have at most six auxiliary vertices. The main new idea is a *negation trick*: since the negation of an assignment is again an assignment, equations with right-hand sides t and $-t$ can be handled together, and this makes the resulting predicates symmetric under all permutations of the colours. We complement it with a general soundness criterion for such reductions: in effect, the best strategy of the adversary is to use a single colour pattern everywhere. This criterion is the strongest analysis that uses Håstad's theorem as a black box, and, using LP duality, we show that under it no gadget reduction from these equations can beat 27/29 for Max-3-Cut.

Finally, we observe that a classical reduction from Max-3-Cut to Max-k-Cut, applied more carefully, already gives $82/85 \approx 0.965$ for Max-4-Cut on 4-colourable graphs, improving the previously stated 85/86.

Contents

1 Introduction **1**
1.1 Our results 1
1.2 Overview of the techniques 2

2 Preliminaries **3**

3 Max-3-Cut: the negation trick **5**

4 A soundness criterion for gadgets from Håstad's equations **6**

5 A uniform family for Max-(3+L)-Cut **8**

6 Universal vertices and perfect completeness **10**

7 A barrier for gadgets from 3Lin(Z3) to Max-3-Cut **11**

8 Discussion **12**

A Methodology and computations **12**

AI methodology **13**

1 Introduction

In Max-k-Cut, we are given a graph with non-negative edge weights and must colour its vertices with k colours so as to maximise the total weight of edges whose endpoints receive different colours. For $k = 2$ this is Max-Cut. For $k \geq 3$ the best approximation algorithms round the semidefinite relaxation of Frieze and Jerrum [FJ97]; they achieve a ratio of about 0.836 for $k = 3$ and 0.857 for $k = 4$ [GW04, dKPW04]. For $k = 3$ this is optimal under the Unique Games Conjecture [IM12, Hei26].

Much less is known about NP-hardness. Kann, Khanna, Lagergren and Panconesi [KKLP97] proved hardness of the form $1 - \Theta(1/k)$, and Guruswami and Sinop [GS13] proved bounds of the same form even for k-colourable graphs. The current record for $k = 3$ was set by Austrin, O'Donnell, Tan and Wright [AOTW14]: using a PCP designed for the purpose, they showed that Max-3-Cut is NP-hard to approximate within $16/17 + \varepsilon$, even on 3-colourable graphs. Combined with a reduction of Guruswami and Sinop, the same result gives the best previously stated bounds for every $k \geq 4$, such as 85/86 for Max-4-Cut [AOTW14]. Most recently, Nagda, Raghavan and Thakurta [NRT25] used the AlphaEvolve system [N$^+$25] to search for gadget reductions from Håstad's linear equations [Hås01]. This improved the bound for Max-4-Cut to about 0.987, but for Max-3-Cut their gadgets give only $55/57 \approx 0.965$, short of 16/17. Table 1 collects these bounds.

1.1 Our results

Our first result breaks the 16/17 barrier for Max-3-Cut.

Theorem 1.1. *For every $\varepsilon > 0$ it is NP-hard, given a weighted graph G, to distinguish between the case that some 3-colouring cuts at least a $(29/30 - \varepsilon)$ fraction of the total edge weight and the case that every 3-colouring cuts at most a $(9/10 + \varepsilon)$ fraction. In particular,* Max-3-Cut *is NP-hard to approximate within $27/29 + \varepsilon \approx 0.931$.*

The same approach, with a few extra vertices added to the gadgets, gives improved bounds for every k.

Theorem 1.2. *For every $k \geq 3$ and $\varepsilon > 0$,* Max-k-Cut *is NP-hard to approximate within $1 - 2/\Gamma_{k-3} + \varepsilon$, where $\Gamma_L = 29 + 18L + 3\binom{L}{2}$. In particular,* Max-4-Cut *is NP-hard to approximate within $45/47 + \varepsilon \approx 0.957$, and* Max-5-Cut *within $33/34 + \varepsilon \approx 0.971$.*

For $k = 3$, Theorem 1.2 is Theorem 1.1. Unlike the result of [AOTW14], our reductions do not have perfect completeness: on 3-colourable graphs, 16/17 remains the best known bound. For k-colourable graphs with $k \geq 4$, however, we observe that the bounds previously stated in the literature can also be improved, by applying a known reduction more carefully.

Theorem 1.3. *For every $\varepsilon > 0$ it is NP-hard to distinguish 4-colourable graphs from graphs in which every 4-colouring cuts at most an $82/85 + \varepsilon \approx 0.965$ fraction of the edge weight, and 5-colourable graphs from graphs in which every 5-colouring cuts at most a $119/122 + \varepsilon \approx 0.975$ fraction. Similar improvements hold for every k from 4 to 20 (Section 6).*

Table 1 compares our bounds with the previous ones. Together, Theorems 1.2 and 1.3 improve the best known NP-hardness of Max-k-Cut for every k from 3 to 20; for larger k the bound of [AOTW14] remains the best. All three theorems have short proofs by hand.

	on k-colourable graphs		on general graphs	
	previous [AOTW14]	Theorem 1.3	previous	Theorems 1.1 and 1.2
$k=3$	$16/17 \approx 0.941$	—	16/17 [AOTW14]	$27/29 \approx 0.931$
$k=4$	$85/86 \approx 0.988$	$82/85 \approx 0.965$	0.987 [NRT25]	$45/47 \approx 0.957$
$k=5$	$120/121 \approx 0.992$	$119/122 \approx 0.975$	120/121 [AOTW14]	$33/34 \approx 0.971$
$k=6$	$101/102 \approx 0.990$	$53/54 \approx 0.981$	101/102 [AOTW14]	$45/46 \approx 0.978$
$k=7$	$136/137 \approx 0.993$	$202/205 \approx 0.985$	136/137 [AOTW14]	$117/119 \approx 0.983$

Table 1: NP-hardness of approximation for Max-k-Cut; smaller factors are stronger. The previous values for $k \geq 4$ are instances of the general formula $1 - \frac{1}{17(k+c_k)+c_k}$, $c_k = k \bmod 3$, of [AOTW14, Thm. 1.2].

1.2 Overview of the techniques

Gadget reductions and the colour-reference problem. Håstad [Hås01] proved that for systems of equations $x_1 + x_2 + x_3 = t$ over $\mathbb{Z}_3$ it is NP-hard to distinguish almost satisfiable systems from systems in which every assignment satisfies at most a $1/3 + \varepsilon$ fraction of the equations. The classical way to turn this into hardness for a graph problem, introduced by Trevisan, Sorkin, Sudan and Williamson [TSSW00] for $\mathbb{Z}_2$ and used for $\mathbb{Z}_3$ by [NRT25], replaces each equation by a small graph, a *gadget*, whose best cut is larger when the equation is satisfied. For $t \neq 0$, however, the predicate "$x_1 + x_2 + x_3 = t$" is not invariant under renaming the colours, so its gadget must refer to a fixed set of *global* vertices that play the role of the colours $0, 1, 2$. In a cheating colouring all global vertices may receive the same colour, and guarding against this makes the gadgets for $t \neq 0$ large and inefficient; in the analysis of [NRT25] they are the bottleneck that keeps the bound above $16/17$.

The negation trick. Our main observation is that the negation $-X$ of an assignment X is again an assignment, and that $-X$ satisfies an equation with right-hand side t exactly when X satisfies the same equation with right-hand side $-t$. If every assignment satisfies at most a $1/3 + \varepsilon$ fraction of the equations, then applying this to both X and $-X$ shows that, for a random equation, the sum s of its variables under X satisfies $s = t$ or $s = -t$ with probability at most about $2/3$. It is therefore enough for the gadgets of the equations with $t = \pm 1$ to detect whether $s \neq 0$, which simply means that exactly two of the three variables are equal. This predicate is invariant under all permutations of the colours, needs no global vertices, and is detected by a triangle together with one extra vertex. Combined with the gadget of [NRT25] for $t = 0$, this proves Theorem 1.1 in a few lines (Section 3).

A soundness criterion. For $k \geq 4$ the satisfying colourings use only three of the k colours, and global vertices become necessary again. We prove a general criterion for gadget reductions from Håstad's equations (Theorem 4.2): the hardness factor equals

$$\rho \;=\; \frac{\max_{x,y} \sum_t V_t(x,y)}{\sum_t C_t},$$

where $V_t(x,y)$ is the value of the gadget for right-hand side t when its three variables receive the colours x and the global vertices the colours y, and C_t is its value on satisfying colourings. In words, a cheating colouring can do no better than to use a single colour pattern x on all equations, and each pattern is charged with the sum of the values of all three gadgets. The proof decodes a

colouring of the graph into many assignments of Håstad's instance and uses a short Fourier-analytic argument; it relies on the fact that Håstad's equations have one variable of one kind and two of another. We also show that no analysis that uses Håstad's theorem only as a black box can do better (Proposition 4.4). The criterion contains the negation trick and, for $k = 3$, the analysis of [NRT25] as special cases. It also turns gadget design into a single linear program over the three gadgets. Solving this program for $k = 4, 5$ and simplifying the solutions by hand led to the gadgets behind Theorem 1.2: the gadgets of Theorem 1.1 together with $k - 3$ global vertices of the extra colours (Section 5).

Universal vertices. Theorem 1.3 uses a reduction of [KKLP97, GS13] that adds vertices joined to the whole graph and forces them to take the extra colours. The previously stated constants apply it on top of a general bound for K-colourable graphs (K divisible by 3) which, for $K = 3$, is three times weaker than the 16/17 bound it is derived from; applying it directly to the 16/17 bound gives Theorem 1.3 (Section 6). Placing the extra vertices inside the gadgets, as in Theorem 1.2, is more economical than adding them to the final graph.

A barrier. Finally, we show that 27/29 is the best factor that the criterion can give for Max-3-Cut, for gadgets of any size (Theorem 7.1): there is a single distribution of colour patterns on which every gadget, for every right-hand side, still achieves at least a 27/29 fraction of its completeness value on average. The proof uses LP duality, in the spirit of the optimality proofs of [TSSW00, HHM$^+$17], with explicit rational certificates that we verify by computer in exact arithmetic; this is the only computer-assisted result of the paper. To go beyond 27/29, one has to change the source problem (for instance, to other predicates covered by Chan's theorem [Cha16]), the form of the reduction, or the analysis.

How these results were obtained. This research was carried out with extensive use of an AI system, which designed and ran the computer searches, found the arguments and drafted the paper under the direction of the author. The methodology, including how every claim was checked, is described in Appendix A and in the AI methodology statement at the end of the paper.

Organisation. Section 2 fixes notation and states Håstad's theorem in the form we use. Section 3 proves Theorem 1.1, Section 4 the soundness criterion, and Section 5 Theorem 1.2. Section 6 treats universal vertices and Theorem 1.3, Section 7 the barrier, and Section 8 open problems.

2 Preliminaries

Cuts. We write $[k] = \{0, 1, \ldots, k-1\}$ and identify $[3]$ with $\mathbb{Z}_3$. A (weighted) graph $G = (V, E, w)$ has non-negative edge weights; for a colouring $\chi\colon V \to [k]$ let $\mathrm{cut}_\chi(G) = \sum_{\{u,v\}\in E} w_{uv}\,[\chi(u) \neq \chi(v)]$ and $w(G) = \sum_e w_e$. We say that Max-k-Cut is (c, s)-hard if it is NP-hard to distinguish $\max_\chi \mathrm{cut}_\chi(G) \geq c\, w(G)$ from $\max_\chi \mathrm{cut}_\chi(G) \leq s\, w(G)$; this implies NP-hardness of approximation within s/c. Weights may be removed at an arbitrarily small loss by the method of [CST01]; see [GS13, Remark 2.6] for Max-k-Cut. The transformation replaces vertices by independent sets and edges by bipartite graphs, so it also preserves k-colourability; hence all results hold for unweighted simple graphs.

Håstad's equations. A *role-separated* instance Φ of $3\textsc{Lin}(\mathbb{Z}_3)$ consists of disjoint sets U and W of variables and a collection of equations e of the form $u_e + w_e + w'_e = t_e$ over $\mathbb{Z}_3$, with $u_e \in U$, distinct $w_e, w'_e \in W$, $t_e \in \mathbb{Z}_3$ (the *type* of e), and weights $\lambda_e \geq 0$ summing to 1. For an assignment $\sigma\colon U \cup W \to \mathbb{Z}_3$, $\mathrm{val}_\Phi(\sigma)$ is the total weight of the equations it satisfies, and $\mathrm{val}(\Phi) = \max_\sigma \mathrm{val}_\Phi(\sigma)$. Φ is *balanced* if each type has total weight $1/3$.

Theorem 2.1 (Håstad [Hås01])**.** *For every $\varepsilon > 0$ it is NP-hard, given a balanced role-separated instance Φ of $3\textsc{Lin}(\mathbb{Z}_3)$, to distinguish $\mathrm{val}(\Phi) \geq 1 - \varepsilon$ from $\mathrm{val}(\Phi) \leq 1/3 + \varepsilon$.*

Comments on the statement. This is Håstad's theorem for the group $\mathbb{Z}_3$ [Hås01, Thm. 5.9]; the gap version follows from Lemmas 5.10 and 5.11 there. In his verifier, each test reads one entry of a folded long-code table indexed by a set U of variables and two entries of a table indexed by a set W of clauses, and accepts iff the three values sum to a constant; folding only changes the right-hand side. Giving the two provers separate tables makes the instance role-separated; the soundness analysis [Hås01, Lemma 5.11] applies to arbitrary tables, i.e. to arbitrary assignments, and constructs the strategies of the two provers from the two kinds of tables separately. Tests whose two W-entries fall into the same folded class have negligible total weight (it tends to 0 with the parameter of the verifier); we discard them, which changes completeness and soundness by a negligible amount.

To balance the types, replace every $u \in U$ by three copies $u^{(0)}, u^{(1)}, u^{(2)}$ and each equation e by the three equations $u_e^{(c)} + w_e + w'_e = t_e + c$ ($c \in \mathbb{Z}_3$), each of weight $\lambda_e/3$. The new instance Φ' is role-separated (with $U \times \mathbb{Z}_3$ and W), and every type has weight exactly $1/3$. An assignment σ of Φ yields $\sigma(u^{(c)}) = \sigma(u) + c$, which satisfies the same weight. Conversely, for any assignment τ of Φ',

$$\mathrm{val}_{\Phi'}(\tau) = \tfrac{1}{3}\textstyle\sum_{c \in \mathbb{Z}_3} \mathrm{val}_\Phi(\sigma_c), \qquad \sigma_c(u) = \tau(u^{(c)}) - c, \quad \sigma_c(w) = \tau(w),$$

and each σ_c is an assignment of Φ (the values on W are shared), so $\mathrm{val}(\Phi') \leq \mathrm{val}(\Phi)$. □

Gadgets. Fix $k \geq 3$. A *gadget* I is a weighted graph on three *primary* vertices p_1, p_2, p_3, k *global* vertices $g_0, \ldots, g_{k-1}$ and some *auxiliary* vertices. For $x \in [k]^3$ and $y \in [k]^k$ let

$$V_I(x, y) = \max\{\mathrm{cut}_\chi(I) : \chi(p_j) = x_j,\ \chi(g_i) = y_i\},$$

the maximum being over the colours of the auxiliary vertices. Let $y^\circ = (0, 1, \ldots, k-1)$ be the canonical colouring of the globals. Since the value of a colouring is invariant under permutations of $[k]$, we may normalise the colours of the globals: let $\mathcal{Y}_k$ be the set of restricted growth strings, i.e. $y \in [k]^k$ whose first entry is 0 and each of whose entries exceeds the maximum of the previous ones by at most 1; every $y \in [k]^k$ is mapped into $\mathcal{Y}_k$ by some permutation of the colours. Since $V_I(\pi x, \pi y) = V_I(x, y)$ for every permutation π of $[k]$, maxima of V_I (or of sums of several V_I) over $y \in \mathcal{Y}_k$ equal the maxima over all $y \in [k]^k$; the normalisation is only a computational saving. A *family* for $3\textsc{Lin}(\mathbb{Z}_3)$ is a triple (I_0, I_1, I_2) of gadgets, I_t to be used for equations of type t. We write $V_t = V_{I_t}$, $S_y(x) = \sum_t V_t(x, y)$, and define the *completeness*

$$C_t = \min\{V_t(x, y^\circ) : x \in \{0,1,2\}^3,\ x_1 + x_2 + x_3 \equiv t \pmod 3\}.$$

Note that the satisfying tuples are coloured with the colours $0, 1, 2$ only.

The reduction. Given a role-separated Φ and a family, the graph $J = J(\Phi)$ has a vertex for every variable of Φ, k global vertices, and for every equation e a copy of I_{t_e} scaled by λ_e, whose primaries p_1, p_2, p_3 are identified with u_e, w_e, w'_e, whose global vertices are identified with the global vertices

of J, and whose auxiliary vertices are new. For balanced Φ, $w(J) = \frac{1}{3}\sum_t w(I_t)$. If $\mathrm{val}(\Phi) \geq 1-\varepsilon$ and Φ is balanced, colouring the variables by an assignment that satisfies weight at least $1-\varepsilon$, the globals by y°, and the auxiliary vertices of each satisfied equation optimally gives

$$\max_{\chi} \mathrm{cut}_\chi(J) \;\geq\; \tfrac{1}{3}\textstyle\sum_t C_t \;-\; \varepsilon \max_t C_t. \tag{1}$$

For every colouring χ of J, after normalising $y = \chi(g_0, \ldots, g_{k-1}) \in \mathcal{Y}_k$,

$$\mathrm{cut}_\chi(J) \;\leq\; \sum_e \lambda_e \, V_{t_e}\big(\chi(u_e), \chi(w_e), \chi(w'_e),\, y\big), \tag{2}$$

since the auxiliary vertices of different copies are disjoint.

3 Max-3-Cut: the negation trick

Here $k = 3$, and the gadgets do not use their global vertices at all. Write $s(x) = x_1 + x_2 + x_3 \in \mathbb{Z}_3$ and observe that $s(x) = 0$ iff the three colours are all equal or all distinct, and $s(x) \neq 0$ iff exactly two of them are equal.

- The *E-gadget* E is the type-0 gadget of [NRT25, Lemma C.1]: auxiliary vertices $a_1, \ldots, a_6$ and the 18 unit-weight edges

  $$p_1a_1,\ p_1a_2,\ p_1a_3,\ p_1a_4,\quad p_2a_3,\ p_2a_4,\ p_2a_5,\ p_2a_6,\quad p_3a_1,\ p_3a_2,\ p_3a_5,\ p_3a_6,$$
  $$a_1a_6,\ a_1a_4,\ a_2a_3,\ a_2a_5,\ a_3a_6,\ a_4a_5.$$

  The auxiliary vertices form the 6-cycle $a_1a_4a_5a_2a_3a_6$, each is adjacent to two primaries, and opposite vertices of the cycle are adjacent to the same pair of primaries; hence E is invariant under all permutations of the primaries.

- The *N-gadget* N (Fig. 1) consists of a unit-weight triangle on p_1, p_2, p_3 and one auxiliary vertex b joined to p_1, p_2, p_3 by edges of weight 3 (total weight 12).

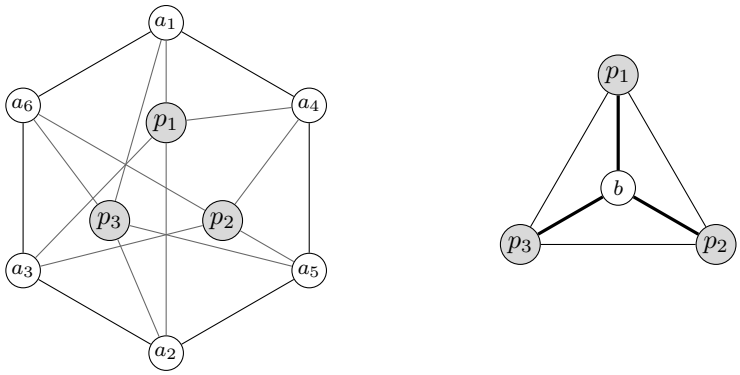
Lemma 3.1. *For all $x \in \mathbb{Z}_3^3$: $V_E(x) = 18$ if $s(x) = 0$ and $V_E(x) = 16$ otherwise; $V_N(x) = 11$ if $s(x) \neq 0$ and $V_N(x) = 9$ otherwise.*

Proof. By symmetry it suffices to consider $x = (a, a, a)$, $x = (a, b, c)$ and $x = (a, a, b)$ with a, b, c distinct. For (a, a, a), colour the cycle alternately with b and c; for (a, b, c), give each a_j the colour of the primary it is not adjacent to (consecutive vertices of the cycle then differ). In both cases all 18 edges are cut. For $x = (a, a, b)$, each of a_1, a_2, a_5, a_6 is adjacent to a primary of colour a and one of colour b, so it has an uncut edge unless it has colour c; since a_1a_6 and a_2a_5 are disjoint edges, at least two edges are uncut. Colouring the cycle $a_1a_4a_5a_2a_3a_6$ by c, b, c, b, c, b leaves exactly p_3a_2 and p_3a_6 uncut, so $V_E(x) = 16$. (This agrees with [NRT25, Lemma C.1], which states completeness 18 and value at most 16 on unsatisfying tuples.)

For N: if exactly two of the x_j are equal, the triangle contributes 2 and b can take the third colour, contributing 9. If all are equal, the triangle contributes 0 and b contributes 9. If all are distinct, the triangle contributes 3 and b agrees with one primary, contributing 6. □

Proof of Theorem 1.1. Use the family $(2E, N, N)$, i.e. E with all weights doubled for type 0 and N for types $1, 2$. By Lemma 3.1, for every x and every type t,

$$V_t(x) \;=\; \beta_t + 2\big([s(x) = t] + [s(x) = -t]\big), \qquad \beta_0 = 32,\ \beta_1 = \beta_2 = 9. \tag{3}$$



(a) E: 18 unit edges (b) N: triangle (weight 1), star (weight 3)

Figure 1: The two gadgets of Theorem 1.1. (a) The type-0 gadget E of [NRT25]: the auxiliary vertices form a 6-cycle, and each is adjacent to two primaries. (b) The gadget N used for types 1 and 2. For $k = 3 + L$ (Section 5) one adds L universal global vertices joined to the auxiliary vertices of E (weight 1), to b and to the primaries in N (weight 3), and to each other inside A_L (weight 3).

Indeed, for $t = 0$ the bracket is $2[s = 0]$, and for $t \neq 0$ it is $[s \neq 0]$. Let Φ be an instance from Theorem 2.1 and $J = J(\Phi)$, so $w(J) = \frac{1}{3}(36 + 12 + 12) = 20$.

Completeness: by (1), $\max_\chi \mathrm{cut}_\chi(J) \geq \frac{1}{3}(36 + 11 + 11) - 36\varepsilon = \frac{58}{3} - 36\varepsilon = (\frac{29}{30} - \frac{9}{5}\varepsilon)\, w(J)$.

Soundness: let χ be any colouring and σ the assignment it induces on the variables of Φ. By (2) and (3),

$$\mathrm{cut}_\chi(J) \leq \sum_e \lambda_e \beta_{t_e} + 2 \sum_e \lambda_e \big([s_e = t_e] + [s_e = -t_e]\big) = \tfrac{50}{3} + 2\big(\mathrm{val}_\Phi(\sigma) + \mathrm{val}_\Phi(-\sigma)\big),$$

where $s_e = \sigma(u_e) + \sigma(w_e) + \sigma(w'_e)$, because $-\sigma$ satisfies e iff $-s_e = t_e$. If $\mathrm{val}(\Phi) \leq \frac{1}{3} + \varepsilon$, both values are at most $\frac{1}{3} + \varepsilon$, so $\mathrm{cut}_\chi(J) \leq \frac{50}{3} + \frac{4}{3} + 4\varepsilon = 18 + 4\varepsilon = (\frac{9}{10} + \frac{\varepsilon}{5})\, w(J)$. □

Remark 3.2. The proof of Theorem 1.1 uses neither role separation nor, essentially, balance: if the type-0 equations have weight w_0 (and types $1, 2$ together $1 - w_0$), the same computation gives the ratio $(23w_0 + \frac{31}{3})/(25w_0 + 11) \leq 31/33 < 16/17$. (For role-separated instances the soundness hypothesis even forces balance in the NO case, see Remark 4.3(c).)

Remark 3.3. In the per-type framework of [NRT25, TSSW00], N is useless as a type-1 gadget: it does not distinguish $s = 1$ from $s = 2$, so its value on unsatisfying inputs can equal its completeness. The gain comes entirely from the analysis, which pairs σ with $-\sigma$.

4 A soundness criterion for gadgets from Håstad's equations

We now prove the general criterion. Throughout, $k \geq 3$ and $\omega = e^{2\pi i/3}$. For maps $a, b\colon [k] \to \mathbb{Z}_3$ define $g_{a,b}\colon \mathbb{Z}_3 \times [k]^3 \to \{0, 1\}$ by

$$g_{a,b}(t, x) = [\, a(x_1) + b(x_2) + b(x_3) = t \,].$$

Lemma 4.1 (Spanning lemma)**.** *Let $f\colon \mathbb{Z}_3 \times [k]^3 \to \mathbb{R}$ satisfy (i) $\sum_{t \in \mathbb{Z}_3} f(t, x) = 0$ for every x, and (ii) $f(t, x_1, x_2, x_3) = f(t, x_1, x_3, x_2)$. Then f is a real linear combination of the functions $g_{a,b} - \frac{1}{3}$.*

Proof. We work over $\mathbb{C}$. For $c \in \mathbb{Z}_3$ let $a + c$ denote the map $v \mapsto a(v) + c$. For fixed (t, x), $g_{a+c,b}(t, x) = 1$ for exactly one c, namely $c = d := t - a(x_1) - b(x_2) - b(x_3)$. Hence for $\ell \in \{1, 2\}$,

$$h_{\ell,a,b} := \sum_{c \in \mathbb{Z}_3} \omega^{\ell c}\big(g_{a+c,b} - \tfrac{1}{3}\big) = \omega^{\ell d} = \omega^{\ell t}\, \omega^{-\ell a(x_1)}\, \omega^{-\ell b(x_2)}\, \omega^{-\ell b(x_3)}.$$

Let $\mathcal{R} \subseteq \mathbb{C}^k$ be the set of vectors whose entries are cube roots of unity. As a and b range over all maps, $\omega^{-\ell a(\cdot)}$ and $\omega^{-\ell b(\cdot)}$ range over all of $\mathcal{R}$ (for both $\ell = 1, 2$). Thus the span contains $\omega^{\ell t} A(x_1)B(x_2)B(x_3)$ for all $A, B \in \mathcal{R}$.

$\mathcal{R}$ spans $\mathbb{C}^k$: it contains $\mathbf{1}$ and $\mathbf{1} + (\omega - 1)e_i$. The tensors $B \otimes B$, $B \in \mathcal{R}$, span the symmetric square $\mathrm{Sym}^2(\mathbb{C}^k)$: if a symmetric matrix Q satisfies $\sum_{i,j} Q_{ij} B_i B_j = 0$ for all $B \in \mathcal{R}$, then writing $B_i = \omega^{\theta_i}$ this says $\sum_{i \le j} Q'_{ij}\, \omega^{\theta_i + \theta_j} = 0$ for all $\theta \in \mathbb{Z}_3^k$, with $Q'_{ii} = Q_{ii}$ and $Q'_{ij} = 2Q_{ij}$ for $i < j$. The functions $\theta \mapsto \omega^{\theta_i + \theta_j}$ $(i \le j)$ are distinct characters of $\mathbb{Z}_3^k$ (their exponent vectors $e_i + e_j$ are distinct since $2 \not\equiv 0, 1 \pmod 3$), hence linearly independent, so $Q = 0$. Consequently the span contains $\omega^{\ell t}\varphi(x_1)\psi(x_2, x_3)$ for every $\varphi \in \mathbb{C}^k$, every symmetric $\psi \in \mathrm{Sym}^2(\mathbb{C}^k)$ and $\ell = 1, 2$.

Now expand $f(t, x) = \sum_{\ell \in \mathbb{Z}_3} \omega^{\ell t} \hat{f}_\ell(x)$ with $\hat{f}_\ell(x) = \frac{1}{3}\sum_t \omega^{-\ell t} f(t, x)$. Condition (i) gives $\hat{f}_0 = 0$, and (ii) makes $\hat{f}_1, \hat{f}_2$ symmetric in (x_2, x_3), i.e. elements of $\mathbb{C}^k \otimes \mathrm{Sym}^2(\mathbb{C}^k)$. So f lies in the complex span of the $g_{a,b} - \frac{1}{3}$; since f and the $g_{a,b}$ are real, taking real parts of the coefficients gives a real combination. □

Theorem 4.2 (Soundness criterion)**.** *Let (I_0, I_1, I_2) be a family of gadgets for* Max-k-Cut*, $k \ge 3$, such that each $V_t(x, y)$ is symmetric under exchanging x_2 and x_3. Let*

$$\rho = \frac{\max_{y \in \mathcal{Y}_k} \max_{x \in [k]^3} S_y(x)}{\sum_{t \in \mathbb{Z}_3} C_t}.$$

Then for every $\delta > 0$, Max-k-Cut *is NP-hard to approximate within $\rho + \delta$. More precisely, with $\bar{w} = \sum_t w(I_t)$,* Max-k-Cut *is $\big(\frac{\sum_t C_t}{\bar{w}} - \delta,\ \frac{\max_{x,y} S_y(x)}{\bar{w}} + \delta\big)$-hard.*

Proof. Reduce from Theorem 2.1 as in Section 2; then $w(J) = \bar{w}/3$. Completeness is (1). For soundness, fix $y \in \mathcal{Y}_k$. The function $f_y(t, x) = V_t(x, y) - \frac{1}{3}S_y(x)$ satisfies (i) and (ii) of Lemma 4.1, so $f_y = \sum_{a,b} \gamma^y_{a,b}(g_{a,b} - \frac{1}{3})$ for some reals $\gamma^y_{a,b}$. Let $K_\gamma = \max_y \sum_{a,b} |\gamma^y_{a,b}|$; it depends only on the family.

Let Φ satisfy $\mathrm{val}(\Phi) \le \frac{1}{3} + \varepsilon$, let χ be a colouring of $J(\Phi)$, and normalise so that the globals have colours $y \in \mathcal{Y}_k$. Let $P(t, x)$ be the total weight of the equations e with $t_e = t$ and $(\chi(u_e), \chi(w_e), \chi(w'_e)) = x$, and let $\bar{P}(x) = \sum_t P(t, x)$. For any maps $a, b\colon [k] \to \mathbb{Z}_3$, the assignment $\sigma_{a,b}(u) = a(\chi(u))$ for $u \in U$ and $\sigma_{a,b}(w) = b(\chi(w))$ for $w \in W$ is well defined because $U \cap W = \emptyset$, and it satisfies exactly the weight $\langle P, g_{a,b}\rangle$. Hence $\langle P, g_{a,b}\rangle \le \frac{1}{3} + \varepsilon$. Since $\sum_c g_{a+c,b} \equiv 1$, also $\langle P, g_{a,b}\rangle = 1 - \sum_{c \ne 0}\langle P, g_{a+c,b}\rangle \ge \frac{1}{3} - 2\varepsilon$. By (2),

$$\mathrm{cut}_\chi(J) \le \sum_{t,x} P(t, x) V_t(x, y) = \sum_x \bar{P}(x)\frac{S_y(x)}{3} + \sum_{a,b} \gamma^y_{a,b}\Big\langle P, g_{a,b} - \tfrac{1}{3}\Big\rangle \le \frac{1}{3}\max_x S_y(x) + 2K_\gamma\varepsilon.$$

Choosing ε small enough compared to δ finishes the proof. □

Remark 4.3. (a) The symmetry hypothesis is harmless: replace I_t by $I'_t = \frac{1}{2}I_t \sqcup \frac{1}{2}I_t^{\mathrm{sw}}$, where I_t^{sw} is I_t with p_2, p_3 exchanged and $\sqcup$ takes disjoint copies of the auxiliary vertices while sharing primaries and globals. Then $V'_t(x, y) = \frac{1}{2}(V_t(x, y) + V_t(x_1, x_3, x_2, y))$, so no C_t decreases (the satisfying sets are symmetric) and $\max_{x,y} S_y(x)$ does not increase. Gadgets that are symmetric under all

permutations of the primaries, as all gadgets in this paper, satisfy the hypothesis trivially. (b) Role separation is essential: if U and W could overlap, only maps with $a = b$ would be available, the shifts $a \mapsto a + c$ would change the sum by $3c = 0$, and the argument breaks down. (c) The proof does not use that the types are balanced in the NO case; with constant maps $a \equiv c_1$, $b \equiv c_2$ the soundness hypothesis in fact forces every type to have weight $\frac{1}{3} \pm O(\varepsilon)$. (d) For $k = 3$, Theorem 4.2 is never worse than the per-type analysis. Every $x \in \mathbb{Z}_3^3$ satisfies exactly one type $t(x)$, so $S_y(x) \le S_1(t(x)) + \sum_{t \neq t(x)} S_2(t)$, where $S_1(t)$ and $S_2(t)$ are the maxima of V_t over satisfying resp. unsatisfying x and all y; hence $\rho \le (\sum_t S_2(t) + \max_t(S_1(t) - S_2(t)))/\sum_t C_t$, which is the bound of the per-type reduction theorem [NRT25, Thm. C.2]. With $S_1(t) = C_t$ and $S_2(t) = C_t - d$ this is $1 - 2/\sum_t \alpha_t$, $\alpha_t = C_t/d$. Theorem 4.2 can be much better because a tuple that is unsatisfying for two types need not be bad for both gadgets simultaneously.

Proposition 4.4. *For every family, every $x^* \in [k]^3$ and every $y^* \in [k]^k$, there is a balanced role-separated instance Φ_0 with $\mathrm{val}_{\Phi_0}(\sigma) = \frac{1}{3}$ for every assignment σ, and a colouring of $J(\Phi_0)$ of value $\frac{1}{3} S_{y^*}(x^*)$. Hence no soundness analysis that uses only the NO conclusion of Theorem 2.1 can improve on the numerator of ρ.*

Proof. Let Φ_0 consist of one variable $u \in U$, two variables $w, w' \in W$ and the three equations $u + w + w' = t$, $t \in \mathbb{Z}_3$, of weight $\frac{1}{3}$ each (or a disjoint union of copies of it). Every assignment satisfies exactly one of the three equations. Colour u, w, w' by x_1^*, x_2^*, x_3^*, the globals by y^*, and the auxiliary vertices optimally. □

Remark 4.5. In terms of the proof of Theorem 4.2: for every x^* the distribution $P(t, x) = \frac{1}{3}[x = x^*]$ satisfies $\langle P, g_{a,b} \rangle = \frac{1}{3}$ for all a, b. Moreover, if the maximum in ρ is attained at a tuple with $x_2^* = x_3^*$, then *every* $J(\Phi)$ has a colouring of value $\frac{1}{3} \max_{x,y} S_y(x)$ (colour all of U by x_1^* and all of W by x_2^*), so the analysis is exact. This is the case for all families in this paper, where the maximum is attained at a constant tuple.

Remark 4.6. The criterion is consistent with the known algorithms: for every family, $\rho \ge \alpha_k$, where α_k is the approximation ratio of the Frieze–Jerrum rounding [FJ97, dKPW04]. Indeed, let $\zeta_0, \dots, \zeta_{k-1}$ be unit vectors with $\langle \zeta_c, \zeta_{c'} \rangle = -\frac{1}{k-1}$ for $c \neq c'$. For each t and each satisfying tuple x of type t (there are 9), fix a colouring χ_x^t of I_t with primaries x, globals y° and value $V_t(x, y^\circ) \ge C_t$, and let $v_z^t = (9^{-1/2} \zeta_{\chi_x^t(z)})_x$ for each vertex z of I_t. These are unit vectors with $\langle v_z^t, v_{z'}^t \rangle = 1 - \frac{k}{k-1} \Pr_x[\chi_x^t(z) \neq \chi_x^t(z')] \ge -\frac{1}{k-1}$, and their Frieze–Jerrum objective $\sum w_{zz'} \frac{k-1}{k}(1 - \langle v_z^t, v_{z'}^t \rangle)$ equals $\mathbb{E}_x V_t(x, y^\circ) \ge C_t$. On the set B of primary and global vertices, the disagreement probabilities do not depend on t (any two primaries are independent and uniform on $\{0, 1, 2\}$), so the Gram matrix of $(v_z^t)_{z \in B}$ is the same for all t. Fix vectors $(f_z)_{z \in B}$ with this Gram matrix, let R_t be the isometry from $\mathrm{span}\{v_z^t : z \in B\}$ onto $\mathrm{span}\{f_z\}$ with $v_z^t \mapsto f_z$, and Π_t the orthogonal projection onto the former span. In a direct sum $\mathbb{R}^m \oplus H_0 \oplus H_1 \oplus H_2$ put $\tilde{v}_z = R_t \Pi_t v_z^t + \iota_t(v_z^t - \Pi_t v_z^t)$ for z in I_t, where ι_t embeds into H_t. Then $\tilde{v}_z = f_z$ on B and all inner products within each I_t are preserved, so this is a vector solution for $I_0 \cup I_1 \cup I_2$ (glued along B) with objective at least $\sum_t C_t$, in which every edge has inner product $\ge -\frac{1}{k-1}$. (Auxiliary vertices of different gadgets are not adjacent, so their inner products do not matter.) The rounding of [FJ97, dKPW04] cuts each such edge with probability at least $\alpha_k \frac{k-1}{k}(1 - \langle \tilde{v}_z, \tilde{v}_{z'} \rangle)$, so some colouring has value at least $\alpha_k \sum_t C_t$; but every colouring has value at most $\max_{x,y} S_y(x)$.

5 A uniform family for Max-(3+L)-Cut

Let $L \ge 0$ and $k = 3 + L$. Call the global vertices $g_3, \dots, g_{k-1}$ (colours $3, \dots, k-1$ in y°) the *universal* vertices $u_1, \dots, u_L$; the globals g_0, g_1, g_2 are not used.

- A_L: the E-gadget of Section 3 (unit weights), plus an edge of weight 1 between every auxiliary vertex a_j and every universal vertex, plus an edge of weight 3 between every pair of universal vertices. Total weight $18 + 6L + 3\binom{L}{2}$.
- N_L: a unit-weight triangle on the primaries, one auxiliary vertex b joined with weight 3 to p_1, p_2, p_3 and to every universal vertex, and an edge of weight 3 between every primary and every universal vertex. Total weight $12 + 12L$.

The family is $(2A_L, N_L, N_L)$. For $L = 0$ it is the family of Section 3.

Lemma 5.1. *For every $L \geq 0$, the gadgets A_L and N_L are invariant under all permutations of the primaries, and the family $(2A_L, N_L, N_L)$ satisfies*

$$C_0 = w(2A_L), \qquad C_1 = C_2 = w(N_L) - 1, \qquad \max_{x,y} S_y(x) = \textstyle\sum_t C_t - 4.$$

Consequently $\sum_t C_t = 2\Gamma_L$ and $\sum_t w(I_t) = 2\Gamma_L + 2$, with $\Gamma_L = 29 + 18L + 3\binom{L}{2}$.

Proof. For a colouring of a gadget, write $\mathrm{col}(v)$ for the colour of a vertex v.

Completeness. Let $x \in \{0,1,2\}^3$ and let the globals have colours y°, so the universal vertices have distinct colours $3, \ldots, k-1$, different from all colours of the primaries. For $t = 0$, colour the a_j as in the proof of Lemma 3.1 (colours in $\{0,1,2\}$): all edges of E, all edges $a_j u_i$ and all edges $u_i u_{i'}$ are cut, so $C_0 = w(2A_L)$. For $t \neq 0$, exactly two primaries are equal; colour b by the third colour of $\{0,1,2\}$: every edge except one triangle edge is cut, so $C_1 = C_2 = w(N_L) - 1$.

Soundness. Fix $x \in [k]^3$ and y, and let $\ell_A = w(A_L) - V_{A_L}(x,y)$ and $\ell_N = w(N_L) - V_{N_L}(x,y)$ be the least uncut weights. Since $S_y(x) = 2V_{A_L} + 2V_{N_L} = \sum_t w(I_t) - 2(\ell_A + \ell_N) = \sum_t C_t + 2 - 2(\ell_A + \ell_N)$, it suffices to show $\ell_A + \ell_N \geq 3$.

In N_L the vertex b is optimally coloured with a colour that is least frequent among its neighbours, so ℓ_N equals the uncut weight of the triangle, plus 3 times the number of pairs (j,i) with $x_j = \mathrm{col}(u_i)$, plus $3 \min_c |\{v \in \{p_1, p_2, p_3, u_1, \ldots, u_L\} : \mathrm{col}(v) = c\}|$. In A_L every pair $u_i u_{i'}$ of equal colour costs 3. Hence we may assume that no primary shares its colour with a universal vertex, that the u_i have distinct colours, and that some colour is missing among the $k = 3 + L$ neighbours of b; otherwise $\ell_A \geq 3$ or $\ell_N \geq 3$. If x is constant, the triangle gives $\ell_N = 3$. If x_1, x_2, x_3 are distinct, then the primaries and the u_i use all k colours, which we excluded. If exactly two coincide, then by symmetry $x = (a, a, b)$; the triangle gives $\ell_N \geq 1$, and exactly one colour c lies outside $\{a, b, \mathrm{col}(u_1), \ldots, \mathrm{col}(u_L)\}$. Each of a_1, a_2, a_5, a_6 is adjacent to a primary of colour a, to a primary of colour b, and to all u_i, so it has an uncut edge unless it has colour c. Since $a_1 a_6$ and $a_2 a_5$ are disjoint edges of E, each of the pairs $\{a_1, a_6\}$ and $\{a_2, a_5\}$ contributes an uncut edge, so $\ell_A \geq 2$. In all cases $\ell_A + \ell_N \geq 3$.

Equality holds for constant x and distinct universal colours different from x_1: then $\ell_N = 3$, and the 6-cycle of E can be coloured with the two colours outside $\{x_1, \mathrm{col}(u_1), \ldots, \mathrm{col}(u_L)\}$, so $\ell_A = 0$. Finally $w(A_L) = 18 + 6L + 3\binom{L}{2}$ and $w(N_L) = 12 + 12L$ give $\sum_t C_t = 2w(A_L) + 2w(N_L) - 2 = 2\Gamma_L$. □

Proof of Theorem 1.2. By Lemma 5.1 and Theorem 4.2 with $\bar{w} = 2\Gamma_L + 2$, Max-k-Cut is $\big(\frac{2\Gamma_L}{2\Gamma_L+2} - \delta, \frac{2\Gamma_L - 4}{2\Gamma_L+2} + \delta\big)$-hard, and $\rho = 1 - 2/\Gamma_L$. For $L = 0, 1, 2, 3, 4$, $\Gamma_L = 29, 47, 68, 92, 119$. □

Remark 5.2. We also confirmed Lemma 5.1 by computer, for $L \leq 6$ (Appendix A, which also lists the values for $L = 1$). Placing the universal vertices inside the gadgets is more economical than adding them to the final graph as in Section 6: combining Theorem 1.1 with Lemma 6.1 gives only 47/49 for Max-4-Cut and 35/36 for Max-5-Cut.

Remark 5.3 (How the family was found). Theorem 4.2 makes the design of a family a single linear program (Appendix A). Seeded with E and N augmented by universal vertices (the construction of Section 6 applied to the gadgets of Section 3), this LP returned the ratios 45/47 and 33/34; we then simplified the optimal solutions into the family above. Our column-generation runs for $k = 4, 5$ did not find better families.

6 Universal vertices and perfect completeness

The following reduction is due to Kann, Khanna, Lagergren and Panconesi [KKLP97, Sec. 3] (for $L = 1$) and Guruswami and Sinop [GS13, Sec. 2.3] (Appendix D of the arXiv version; for $L \le 2$, where the base graph is K-colourable with $3 \mid K$ and the weights are d_v/K).

Lemma 6.1. *Let G be a weighted graph with total weight M and weighted degrees d_v, let $L \ge 1$ and $w \ge 0$. Let H be obtained by adding vertices $u_1, \dots, u_L$, joining each u_i to every $v \in V(G)$ with weight $d_v/3$, and joining every pair u_i, u_j with weight w. For a graph G' let $\mathrm{unc}_k(G')$ be the minimum weight of monochromatic edges over k-colourings of G'. Then*

$$\min\{w,\ \mathrm{unc}_3(G)\} \ \le\ \mathrm{unc}_{3+L}(H) \ \le\ \mathrm{unc}_3(G),$$

and $w(H) = M(1 + \frac{2L}{3}) + \binom{L}{2} w$. If G is 3-colourable then H is $(3+L)$-colourable.

Proof. The upper bound: give $u_1, \dots, u_L$ the colours $3, \dots, L+2$ and colour G optimally with $\{0, 1, 2\}$. For the lower bound, let χ be a $(3+L)$-colouring of H. If two u_i share a colour, χ loses at least w. Otherwise the u_i use L distinct colours, and a vertex $v \in V(G)$ can share its colour with at most one u_i. If it does, the edge vu_i is monochromatic, of weight $d_v/3$. Among the three colours not used by the u_i, some colour is used by neighbours of v of total weight at most $d_v/3$; recolouring v with it does not increase the weight of monochromatic edges. Repeating this for all such v yields a colouring that uses only three colours on G, whose monochromatic weight is at least $\mathrm{unc}_3(G)$. The total weight follows from $\sum_v d_v = 2M$. □

If G comes from a (c, s)-hard instance of Max-3-Cut with $\delta_c = 1 - c$ and $\delta_s = 1 - s$, choose $w = \delta_s M$. Then $\mathrm{unc}_{3+L}(H) \le \delta_c M$ in the YES case and $\ge \delta_s M$ in the NO case, so Max-3+L-Cut is hard within

$$\frac{m_L - \delta_s}{m_L - \delta_c}, \qquad m_L = 1 + \tfrac{2L}{3} + \tbinom{L}{2}\delta_s. \tag{4}$$

Proof of Theorem 1.3. Apply (4) to the $(1, \frac{16}{17} + \varepsilon)$-hardness of 3-colouring of [AOTW14] ($\delta_c = 0$, $\delta_s = \frac{1}{17}$): the ratio is $1 - \frac{\delta_s}{m_L} = 1 - \frac{3}{51 + 34L + 3\binom{L}{2}}$, e.g. $\frac{82}{85}$ for $L = 1$ and $\frac{119}{122}$ for $L = 2$. Perfect completeness is preserved. □

Remark 6.2. Where do the previously stated constants come from? Guruswami and Sinop first prove a bound for K-colourable graphs, for every K divisible by 3, by a tensor-product construction, and then add $k - K \le 2$ universal vertices [GS13, Sec. 2.3]. Their bound for K colours loses a factor K in the gap compared with the bound for 3 colours that it is built from, and they use it also for $K = 3$, where this loss is unnecessary; [AOTW14] plugged their 16/17 into the resulting formula. (For $k = 3$ the formula gives only 50/51.) Starting from the 3-colour bound directly, (4) improves on the formula for every k from 4 to 20; even the 32/33 of [GS13] would give 54/55 for Max-4-Cut.

7 A barrier for gadgets from 3Lin(Z3) to Max-3-Cut

Optimality proofs for gadgets via LP duality go back to [TSSW00]; [HHM$^+$17] proved an analogous optimality statement for gadgets from Chan's Hadamard predicates to 2Lin($\mathbb{Z}_2$).

Theorem 7.1 (computer-assisted)**.** *For every family (I_0, I_1, I_2) of* Max-3-Cut *gadgets (with any number of auxiliary vertices, possibly using the global vertices),*

$$\max_{x \in \mathbb{Z}_3^3} S_{y^\circ}(x) \;\geq\; \tfrac{27}{29} \textstyle\sum_t C_t.$$

Hence Theorem 4.2 cannot give a ratio below 27/29 for Max-3-Cut*, and the family of Section 3 is optimal for it.*

The proof rests on lower bounds for single gadgets against a fixed distribution of inputs, which are instances of weak LP duality.

Lemma 7.2. *Fix t and let $\mathcal{F}_t = [3]^{\mathrm{sat}_t}$ be the set of* columns *(functions from $\mathrm{sat}_t = \{x \in \mathbb{Z}_3^3 : s(x) = t\}$ to colours). Suppose $\theta \colon \mathrm{sat}_t \to \mathbb{R}_{\geq 0}$ and a probability distribution ν over pairs $r = (x_r, \chi_r)$, with $x_r \in \mathbb{Z}_3^3$ and χ_r a colouring of the* universe $\Omega = \{p_1, p_2, p_3, g_0, g_1, g_2\} \cup \mathcal{F}_t$ *with $\chi_r(p_j) = (x_r)_j$ and $\chi_r(g_i) = i$, satisfy, for every pair $\{\xi, \eta\} \subseteq \Omega$,*

$$\Pr_{r \sim \nu}[\chi_r(\xi) \neq \chi_r(\eta)] \;\geq\; \sum_{x \in \mathrm{sat}_t} \theta(x)\,[\mathrm{can}_x(\xi) \neq \mathrm{can}_x(\eta)], \tag{5}$$

where can_x is the canonical colouring of Ω at x: $\mathrm{can}_x(p_j) = x_j$, $\mathrm{can}_x(g_i) = i$, $\mathrm{can}_x(f) = f(x)$. Then every type-t gadget satisfies $\mathbb{E}_{r \sim \nu} V_t(x_r, y^\circ) \geq \big(\sum_x \theta(x)\big)\, C_t$.

Proof. For each $x \in \mathrm{sat}_t$ fix a colouring of the auxiliary vertices of I_t attaining $V_t(x, y^\circ) \geq C_t$. Let $\phi \colon V(I_t) \to \Omega$ map p_j and g_i to themselves and each auxiliary vertex to its column (its colour as a function of x). Then $\mathrm{can}_x \circ \phi$ is the fixed optimal colouring at x, and $\chi_r \circ \phi$ is a colouring with primaries x_r and globals y°, of value at most $V_t(x_r, y^\circ)$. Summing (5) over the edges $\xi'\eta'$ of I_t with $\phi(\xi') \neq \phi(\eta')$, with their non-negative weights, gives $\mathbb{E}_r V_t(x_r, y^\circ) \geq \sum_x \theta(x) V_t(x, y^\circ) \geq (\sum_x \theta(x))\, C_t$; edges with $\phi(\xi') = \phi(\eta')$ are never cut by $\mathrm{can}_x \circ \phi$. □

Proof of Theorem 7.1. Let $\mathcal{A}$ (all equal, 3 tuples), $\mathcal{D}$ (all distinct, 6 tuples), $\mathcal{T}_1$ and $\mathcal{T}_2$ (two equal with sum 1, resp. 2; 9 tuples each) partition $\mathbb{Z}_3^3$, so that $\mathrm{sat}_0 = \mathcal{A} \cup \mathcal{D}$. We use certificates as in Lemma 7.2 (Appendix A):

- type 0: x_r uniform on $\mathcal{T}_1 \cup \mathcal{T}_2$ and $\sum \theta = \frac{8}{9}$;
- type 1: $x_r \in \mathcal{A}, \mathcal{D}, \mathcal{T}_2$ with probabilities $\frac{5}{20}, \frac{6}{20}, \frac{9}{20}$, uniform within each class, and $\sum \theta = \frac{9}{10}$;
- type 2: for a type-2 gadget I, let I' be I with g_1 and g_2 exchanged. By invariance under the colour permutation $c \mapsto -c$, $V_{I'}(x, y^\circ) = V_I(-x, y^\circ)$, so I' is a type-1 gadget with the same completeness, and the type-1 certificate gives $\mathbb{E}\, V_I(-x_r, y^\circ) \geq \frac{9}{10} C_I$, where $-x_r \in \mathcal{A}, \mathcal{D}, \mathcal{T}_1$ with probabilities $\frac{5}{20}, \frac{6}{20}, \frac{9}{20}$, uniform within each class. (We also stored and checked this mirrored certificate explicitly.)

In each certificate every row has $x_r \notin \mathrm{sat}_t$. Let μ^* put mass $\frac{5}{29}$ uniformly on $\mathcal{A}$ and mass $\frac{1}{29}$ on each tuple of $\mathcal{D} \cup \mathcal{T}_1 \cup \mathcal{T}_2$. Then, since $V_t(x, y^\circ) \geq C_t$ on sat_t,

$$\mathbb{E}_{\mu^*} V_0 \geq \tfrac{11}{29} C_0 + \tfrac{18}{29} \cdot \tfrac{8}{9} C_0 = \tfrac{27}{29} C_0, \qquad \mathbb{E}_{\mu^*} V_1 \geq \tfrac{9}{29} C_1 + \tfrac{20}{29} \cdot \tfrac{9}{10} C_1 = \tfrac{27}{29} C_1,$$

and likewise for V_2. Summing over t, $\max_x S_{y^\circ}(x) \geq \mathbb{E}_{\mu^*} S_{y^\circ}(x) \geq \frac{27}{29} \sum_t C_t$. □

Remark 7.3. In each row r of the certificates, the colour of a column f is $h_r(f|_{T_r})$ for a set T_r of one or two satisfying tuples and a table $h_r\colon [3]^{T_r} \to [3]$. The certificates were found by solving the dual LP with constraint generation over pairs, and then rounded to rationals: they have 78, 162 and 162 rows of positive weight and common denominators 324, 4680 and 4680. The rounded certificates are what is verified: (5) was checked *exactly*, for all pairs of elements of Ω ($|\mathcal{F}_t| = 3^9$), by two programs using different methods; the minimum slack is 0 (Appendix A).

8 Discussion

- *Perfect completeness.* Our reductions lose completeness (29/30); on 3-colourable graphs 16/17 [AOTW14] is still the best bound (the first constant bound there is due to [Pet94]). Can the negation trick be combined with a source that has perfect completeness?

- *Other sources.* By Theorem 7.1 and Proposition 4.4, per-type gadgets on the instances of Theorem 2.1, analysed through Theorem 2.1 alone, cannot beat 27/29. Natural candidates for other sources are the predicates $\sum_i a_i x_i = t$ over $\mathbb{Z}_3$ with $a \in \{\pm 1\}^r$ and $\sum_i a_i \equiv 0$, which are NP-hard to approximate by Chan's theorem [Cha16] (the unconditional analogue of [AM09]) and for which the negation trick again yields colour-symmetric predicates; gadget reductions from Chan's predicates have been used successfully for 2Lin($\mathbb{Z}_2$) [HHM$^+$17]. We expect the proof of Theorem 4.2 to carry over to Chan's r-partite instances, with one decoding map per player (the analogue of Lemma 4.1 for products of r arbitrary vectors), in which case no symmetry hypothesis would be needed.

- *Optimality for $k \geq 4$.* We do not know whether 45/47 and 33/34 are optimal under Theorem 4.2; our column generation found nothing better, but we have no barrier certificate.

- *Large k.* Our family gives $1 - \Theta(1/k^2)$; combining Theorem 4.2 with the tensor construction of [GS13] might improve the constant in $1-\Theta(1/k)$ and the range $k \leq 20$ in which we improve the known bounds.

A Methodology and computations

How the results were obtained. The research was carried out with an AI system (Claude, by Anthropic) working in an agentic coding environment, under the direction of the author, who set the goals at each stage (first improving the gadgets of [NRT25] for Max-4-Cut and Max-5-Cut, then attacking Max-3-Cut) and decided which directions to pursue. The AI system wrote and ran the linear-programming searches for gadgets and the verification programs, proposed the negation trick of Section 3 while exploring alternative source problems, derived the soundness criterion of Section 4, found the gadget families of Section 5 (by the linear program described below, followed by simplification) and the hand proof of Lemma 5.1, observed the consequences of the universal-vertex reduction in Section 6, computed the certificates of Section 7, searched the literature, and drafted the text. Every main claim was checked in two ways. First, independently instantiated AI agents, which did not see the reasoning that produced a claim, were asked to refute it: they re-read the primary sources (in particular Håstad's verifier and the constants of [GS13, AOTW14]), re-derived the proofs, and re-implemented the computations from scratch; four such referees also reviewed a draft of this paper. Second, every gadget value and every certificate was verified by at least two independently written programs, all using exact arithmetic, as described below.

Availability. The code, the gadget families and the certificates are available at `https://github.com/ziangni-sys/max-3-cut`. All programs are in Python 3 with NumPy; HiGHS was used for the linear programs.

Verification of families. `scripts/verify_sum_criterion.py`, applied to the families produced by `scripts/make_unified_family.py`, computes $V_t(x, y)$ for all $x \in [k]^3$ and $y \in \mathcal{Y}_k$ by exhaustive enumeration of the auxiliary colourings in integer arithmetic, checks the symmetry hypothesis of Theorem 4.2, and outputs the exact ratio ρ; it reproduces Lemma 5.1 for $L \le 3$. The independently written `scripts/independent/family_dp.py` (which optimises the auxiliary vertices of A_L by dynamic programming along the 6-cycle) reproduces it for $L \le 6$. For $L = 1$ the values (V_{2A_1}, V_{N_1}), by the pattern of equalities among $(x_1, x_2, x_3 \,|\, \mathrm{col}(u_1))$, are: $aaa|a$: $(48, 12)$; $aaa|b$: $(48, 21)$; $aab|a$: $(48, 17)$; $aab|b$: $(48, 20)$; $aab|c$: $(44, 23)$; $abc|a$ and $abc|d$: $(48, 21)$, so $S = 72, 90, 82, 88, 90, 90$; the remaining patterns follow by permuting the primaries. `scripts/verify_sym3.py` checks Lemma 3.1 and the identity (3); the end-to-end reduction of Section 3 was also tested on random small instances of $3\textsc{Lin}(\mathbb{Z}_3)$ by brute force.

The design LP. For fixed candidate auxiliary vertices, each described by its column (its colour in the completeness colouring at each satisfying tuple), each type-t block has one variable per orbit of edges under permutations of the primaries. Completeness rows require the canonical colouring at each satisfying tuple to have value at least c_t; soundness is enforced by variables $v_t(x, y)$ with rows $v_t(x, y) \ge$ (value of an explicit colouring), generated by local search and verified by exact separation, and coupling rows $\tau \ge \sum_t v_t(x, y)$ for all (x, y); we minimise τ subject to $\sum_t c_t \ge 1$. New auxiliary vertices were generated by exact dual pricing of single vertices (over all 4^9 columns for $k = 4$) and by pricing whole small gadgets with up to 7 auxiliary vertices; in our runs the optimum stayed at $45/47$ ($k = 4$) and $33/34$ ($k = 5$).

Barrier certificates. The certificates of Theorem 7.1 are stored as exact integers (common denominator D, row weights, θ, and for each row x_r, T_r, h_r) in the files `cert_t0.json`, `cert_t1.json` and `cert_t2.json` of the directory `results/barrier_k3`. The checker `check_exact.py` verifies that $\theta \ge 0$, that ν is a probability distribution whose marginal on x_r is uniform within each class, that $\sum \theta$ equals $\frac{8}{9}$ resp. $\frac{9}{10}$, and verifies (5) for all pairs of elements of Ω, including all $\binom{3^9}{2}$ pairs of columns, with integer-valued floating-point matrix products whose partial sums stay below 2^{53} (hence exact). A second, independently written checker (`check_exact_pairwise.py`) writes the slack of a pair of columns (f, g) as a sum of pairwise terms in $z_s = (f(s), g(s))$, $s \in \mathrm{sat}_t$, and minimises it exactly over all 9^9 values. Both find minimum slack 0.

Universal vertices. `scripts/universal_vertex.py` checks Lemma 6.1 by brute force on random weighted graphs and computes (4).

AI methodology

We used Claude (Anthropic), a large language model, in an agentic coding environment (Claude Code, including multi-agent workflows), to carry out this research and to write this paper. The tool proposed the main ideas (the negation trick, the soundness criterion and its proof, the gadget families and the proof of Lemma 5.1, the use of the universal-vertex reduction, and the barrier certificates), wrote and ran all search and verification programs, performed the literature search, and drafted the text of all sections. It therefore materially affected all sections of the paper; the

methodology, including how the claims were checked by independent AI agents and by independent programs, is described in Appendix A. The author directed the research and is responsible for the content of the submission. The author verified the correctness and originality of all content, including references.

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